Why Samsung is Losing the Chip War

【三星为何输掉芯片战争 20250713】

Summary: Samsung, once the leader in memory and logic chips, faces declining market share in foundry and memory due to yield issues, lost customers, and fierce competition from TSMC and SK Hynix.

摘要： 三星曾是内存和逻辑芯片的领导者，但由于良率问题、客户流失以及台积电和SK海力士的激烈竞争，其在代工和内存领域的市场份额不断下滑。

⏱ Estimated Reading Time: 36 min

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Imagine being the undisputed King of Memory chips for decades, only to see your crown slipping away in just a few short years.

想象一下，作为几十年来无可争议的内存芯片之王，却在短短几年内眼睁睁看着王冠滑落。

What if the tech giant that powers your smartphones and TVs is suddenly on the back foot in the chip race?

如果为你的智能手机和电视提供芯片的科技巨头突然在芯片竞赛中处于劣势，会怎样?

Today we're diving into a looming tech shake-up.

今天，我们将深入探讨一场即将到来的科技变革。

What happened to Samsung's semiconductor empire?

三星的半导体帝国发生了什么?

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Samsung Electronics, the pride of South Korea and once a dominating force in both memory and logic chips, is facing one of its toughest moments.

三星电子，韩国的骄傲，曾经在内存和逻辑芯片领域占据主导地位，如今正面临最严峻的时刻之一。

Its stock has plunged as investors fear it's losing out to small arrival SK high-nix in AI memory and failing to gain on TSMC in outsourced chip making.

由于投资者担心三星在AI内存领域输给新秀SK海力士，并在外包芯片制造上未能赶上台积电，其股价暴跌。

In fact, Samsung's market value dropped by $122 billion in late 2024 amid these concerns.

事实上，由于这些担忧，三星的市值在2024年底缩水了1220亿美元。

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Just months earlier, profits were soaring and optimism was high.

就在几个月前，利润还在飙升，乐观情绪高涨。

Now it's become a cautionary tale of how quickly fortunes can turn in the chip industry.

如今，这已成为一个警示故事，说明芯片行业的命运可以多么迅速地逆转。

But why is Samsung struggling now?

但为什么三星现在陷入困境?

And can they stage a comeback?

他们能否卷土重来?

Or is this the start of a long decline?

或者这是长期衰落的开始?

Let's explore the rise and stumble of Samsung's semiconductor ambitions.

让我们探讨三星半导体雄心的崛起与挫折。

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To understand Samsung's current troubles, we need context.

要理解三星当前的困境，我们需要背景信息。

Samsung's semiconductor journey began in earnest in the 1980s when it boldly entered the memory chip business.

三星的半导体之旅真正始于1980年代，当时它大胆进军内存芯片业务。

By the 1990s, Samsung had become the world's top manufacturer of memory, DRAM, the working memory in computers, and later NAND Flash, the storage in SSDs and phones.

到了1990年代，三星已成为全球顶级内存制造商，生产计算机的工作内存DRAM，以及后来用于SSD和手机的存储芯片NAND Flash。

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These memory chips were Samsung's golden goose, pumping billions in profits during tech booms.

这些内存芯片是三星的金鹅，在科技繁荣时期带来数十亿美元的利润。

In the 2010s, memory was so lucrative that Samsung's chip division became the company's biggest profit center, even more than smartphones during certain years.

在2010年代，内存利润如此丰厚，以至于三星的芯片部门成为公司最大的利润中心，在某些年份甚至超过智能手机业务。

But Samsung wasn't content with just memory.

但三星并不满足于仅做内存。

It had its eyes on a bigger prize, logic chips, the processors and SOCs, systems on chip that act as brains of devices.

它将目光投向更大的目标——逻辑芯片，即作为设备大脑的处理器和片上系统（SOC）。

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This led Samsung to launch its Foundry division, offering to manufacture chips for external clients much like Taiwan's TSMC does.

这促使三星成立代工部门，像台积电一样为外部客户制造芯片。

Samsung's ambition was clear.

三星的野心很明确。

It wanted to rival TSMC in Intel and become the world's leader in all semiconductor areas.

它希望与台积电和英特尔竞争，成为全球半导体领域的领导者。

The company announced a massive investment plan of 171 trillion won, about $151 billion by 2030 to achieve this goal.

该公司宣布了一项庞大的投资计划，到2030年投入171万亿韩元（约1510亿美元）以实现这一目标。

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This war chest was meant to fund cutting-edge research and huge new fabs, chip factories.

这笔资金旨在资助尖端研究和建设大型新晶圆厂。

Samsung proudly touted its Vision 2030 to topple TSMC in Foundry and reinforce its memory dominance.

三星自豪地宣扬其“2030愿景”，旨在在代工领域击败台积电并巩固其内存主导地位。

For a while, things looked promising.

一段时间内，形势看起来很有希望。

Samsung was an early adopter of extreme ultraviolet, EUV lithography, a next-gen chipmaking technology.

三星是极紫外光刻（EUV）这一下一代芯片制造技术的早期采用者。

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In fact, Samsung was the first to use EUV for seven nanometer chips in 2018, beating even TSMC to that milestone.

事实上，三星在2018年率先将EUV用于7纳米芯片，甚至比台积电更早达到这一里程碑。

Technical aside, seven nanometers, five nanometers, three nanometers, these numbers refer to successive generations of fabrication technology.

技术上讲，7纳米、5纳米、3纳米这些数字指的是连续的制程技术世代。

Smaller node numbers generally mean more transistors packed in, which usually improves efficiency and power.

更小的节点数字通常意味着可以塞入更多晶体管，从而提高效率和性能。

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But as these numbers shrink to mere billions of a meter, making chips becomes astronomically complex.

但随着这些数字缩小到仅几纳米，芯片制造变得极其复杂。

Samsung also pioneered a new transistor architecture called Gate All Around, GAA.

三星还率先推出了一种名为全环绕栅极（GAA）的新型晶体管架构。

In 2022, Samsung announced it had started mass production of a three nanometer GAA process, again, ahead of TSMC.

2022年，三星宣布开始量产3纳米GAA工艺，再次领先于台积电。

GAA is an advanced design where the transistors channel is surrounded by the gate on all sides, as opposed to just three sides in the older FinFET design.

GAA是一种先进设计，晶体管的沟道被栅极四面环绕，而不是像旧式FinFET设计那样仅三面环绕。

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In simple terms, GAA can reduce leakage and improve control at ultra small scales, potentially delivering better performance and efficiency.

简单来说，GAA可以减少漏电并在超小尺度上提高控制力，从而可能提供更好的性能和效率。

This was supposed to be Samsung's trump card to leapfrog TSMC technologically.

这原本是三星在技术上超越台积电的王牌。

Samsung's early foundry efforts even scored some big-name clients.

三星早期的代工努力甚至赢得了一些大客户。

Apple dual sourced its A9 iPhone chips from Samsung at 14 nanometers, and TSMC at 16 nanometers in 2015, and Samsung had bragging rights for making iPhone brains.

2015年，苹果从三星（14纳米）和台积电（16纳米）双源采购A9 iPhone芯片，三星曾为制造iPhone的“大脑”而自豪。

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Around 2016, Qualcomm tapped Samsung's fabs to produce flagship Snapdragon processors on 14 nanometers and 10 nanometers.

2016年左右，高通选择三星的晶圆厂生产14纳米和10纳米的旗舰骁龙处理器。

In video two, used Samsung's 8 nanometer process for its G4RTX 3000 series GPUs in 2020.

2020年，英伟达使用三星的8纳米工艺生产其RTX 3000系列GPU。

These wins signaled that Samsung could compete with TSMC on advanced chip manufacturing.

这些胜利表明，三星可以在先进芯片制造领域与台积电竞争。

Samsung also, of course, manufactured its own X-Inos processors in-house.

当然，三星还自行生产其Exynos处理器。

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All told, by the late 2010s, Samsung had become a solid number two in the Foundry business behind TSMC while enjoying an unchallenged number one spot in memory.

总的来说，到2010年代末，三星已成为代工业务中仅次于台积电的稳固第二，同时在内存领域占据无可争议的第一。

It truly seemed like a semiconductor empire.

这看起来确实像一个半导体帝国。

Memory dominance plus rising foundry prowess.

内存主导地位加上不断增长的代工实力。

However, in this high stakes tech arena, today's leader can become tomorrow's laggard if they miss a beat.

然而，在这个高风险的科技竞技场中，今天的领导者如果错过一步，就可能成为明天的落后者。

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Samsung's lofty goals and initial successes masked underlying cracks that have since widened.

三星的崇高目标和初步成功掩盖了后来扩大的潜在裂痕。

In the world of chip manufacturing, Taiwan's TSMC has long been the juggernaut, a pure play foundry serving the likes of Apple, AMD, Nvidia, and more.

在芯片制造领域，台积电长期以来一直是巨头，是一家纯粹的代工厂，为苹果、AMD、英伟达等公司服务。

Samsung, with all its resources, aimed to challenge this dominance.

三星凭借其所有资源，旨在挑战这一主导地位。

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Yet today, the gap between TSMC and Samsung is only growing.

然而今天，台积电和三星之间的差距只会越来越大。

Just look at the market share.

只需看看市场份额。

TSMC now controls roughly 67% of the global Foundry market while Samsung's share has dwindled to single digits.

台积电现在控制着全球代工市场约67%的份额，而三星的份额已降至个位数。

In early 2025, Samsung's Foundry market share was estimated at only 7.7%.

2025年初，三星的代工市场份额估计仅为7.7%。

A steady slide down from 8.1% the previous quarter and much higher percentages a few years ago.

从上一季度的8.1%稳步下滑，远低于几年前的更高比例。

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For context, Samsung had around approximately 15% share a few years prior.

作为背景，三星几年前的市场份额约为15%。

It's basically been cut in half as it loses ground.

随着其失去优势，份额基本上减半。

Samsung is still the second largest foundry, but it's a very distant number two, and even that position is at risk.

三星仍然是第二大代工厂，但远远落后于第一，甚至这一地位也岌岌可危。

China's SMIC, a rising competitor, reached a 6.0% share and is nipping at Samsung's heels.

中国的竞争对手中芯国际（SMIC）已达到6.0%的份额，正在紧追三星。

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In other words, Samsung is now barely holding on to the number two spot.

换句话说，三星现在勉强保持着第二的位置。

In a market it once hoped to lead.

在一个它曾经希望领导的市场中。

How did this happen?

这是怎么发生的?

Rhetorical question.

反问句。

How did Samsung's Foundry dream stumble despite its deep pockets and early tech leaps?

尽管资金雄厚且早期技术领先，三星的代工梦想为何受挫?

There isn't one simple answer.

没有一个简单的答案。

Rather a combination of technology challenges, customer trust issues, and fierce competition.

而是技术挑战、客户信任问题和激烈竞争的综合结果。

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Let's break it down.

让我们分解一下。

Technology and yield struggles.

技术和良率问题。

Cutting edge chip manufacturing is unforgiving.

尖端芯片制造是残酷的。

While Samsung beat TSMC to 3 nanometer GAA technology on paper, it's struggled to get good yields, i.e. a high percentage of working chips per wafer on those advanced nodes.

虽然三星在纸面上比台积电更早实现3纳米GAA技术，但其良率（即每片晶圆上可用芯片的比例）一直难以提高。

In 2024, reports emerged that Samsung's 3 nanometer GAA process yields were painfully low, and this scared away potential clients.

2024年，有报道称三星的3纳米GAA工艺良率极低，这吓跑了潜在客户。

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An example, Samsung's own X-NOS 2500 chip, planned to use the 3 nanometer node, was reportedly yielding so poorly that it cast doubt on whether it could even be used in Samsung's next Galaxy S phone.

例如，三星自己的Exynos 2500芯片计划使用3纳米节点，但据报道良率极差，甚至让人怀疑它能否用于三星下一款Galaxy S手机。

And if Samsung can't get good yields for itself, imagine the hesitation of outside customers.

如果三星自己都无法获得良好的良率，想象一下外部客户的犹豫。

High performance chip designers like Apple, Nvidia, and AMD Prize, not just the theoretical capability of a process, but its reliability and efficiency in mass production.

苹果、英伟达和AMD等高性能芯片设计公司不仅看重工艺的理论能力，还看重其在大规模生产中的可靠性和效率。

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TSMC has earned a reputation for excellent yields and on-time delivery at each node, five nanometers, four nanometers, three nanometers, and so on, whereas Samsung hit some bumps.

台积电以在每个节点（5纳米、4纳米、3纳米等）上出色的良率和准时交付而闻名，而三星则遇到了一些问题。

Even at the four nanometer, five nanometer generation, there were murmurs that Samsung's processes had lower yields or higher power consumption.

即使在4纳米、5纳米世代，也有传言称三星的工艺良率较低或功耗较高。

In fact, Qualcomm, which used Samsung's four nanometers for the Snapdragon 8 Gen 1 mobile processor, saw that chip plagued by thermal issues, the improved Snapdragon 8 Plus and subsequent Gen 2 moved to TSMC's process, yielding better power efficiency.

事实上，高通在骁龙8 Gen 1移动处理器上使用了三星的4纳米工艺，但该芯片饱受发热问题困扰，改进版的骁龙8 Plus和随后的Gen 2转向台积电工艺，获得了更好的能效。

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These kinds of shifts were bad optics for Samsung foundry's tech image.

这类转变对三星代工的技术形象不利。

Samsung's bleeding edge innovation also perhaps got ahead of its execution.

三星的尖端创新可能也超出了其执行能力。

They were first to attempt GAA transistors, which are indeed the future, but TSMC stuck to a more mature finfet for three nanometers and made it work well.

三星率先尝试GAA晶体管，这确实是未来的方向，但台积电坚持使用更成熟的FinFET工艺用于3纳米并取得了良好效果。

TSMC's customers seemingly preferred a reliable advanced finfet over Samsung's early GAA with issues.

台积电的客户似乎更喜欢可靠的先进FinFET，而不是三星有问题的早期GAA。

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As a result, TSMC's three nanometer, called N3, is thriving.

因此，台积电的3纳米工艺（称为N3）蓬勃发展。

It made up 26% of TSMC's wafer sales by Q4 2024, thanks to Apple's iPhone chips primarily, whereas Samsung's three nanometer had no major commercial customer at that time.

到2024年第四季度，它占台积电晶圆销售额的26%，主要得益于苹果的iPhone芯片，而三星的3纳米工艺当时没有主要商业客户。

Yield rate matters immensely.

良率极其重要。

A cutting-edge fab costs tens of billions of dollars, and if only say 20 to 30% of chips per batch are sellable, profitability is impossible.

一座尖端晶圆厂耗资数百亿美元，如果每批芯片中只有20%到30%是可销售的，就不可能盈利。

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Samsung appears to have learned this the hard way, with insiders suggesting its foundry business has posted consecutive losses due to these challenges.

三星似乎已经艰难地吸取了这一教训，内部人士表示，由于这些挑战，其代工业务连续亏损。

By one estimate, Samsung's foundry and logic design, System LSI unit, lost roughly 500 billion won, about $385 million, in the third quarter of 2024 alone, a stark contrast to TSMC's hefty profits in the same period.

据估计，仅2024年第三季度，三星的代工和逻辑设计（系统LSI部门）就亏损约5000亿韩元（约3.85亿美元），与台积电同期的高额利润形成鲜明对比。

In fact, Samsung even reportedly delayed bringing in some brand new EUV lithography machines to its upcoming Texas fab, because it hasn't secured enough customer orders yet to use them.

事实上，据报道，三星甚至推迟了将一些全新的EUV光刻机引入其即将建成的德州晶圆厂，因为它尚未获得足够的客户订单来使用它们。

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That's right, the company is literally holding off on installing cutting-edge equipment, like ASMR's famous EUV scanners, because demand isn't there.

没错，该公司实际上推迟安装尖端设备，如ASML著名的EUV扫描仪，因为需求不足。

It's an almost unthinkable situation for a firm that wanted to fill fabs, not idle them.

对于一家希望填满晶圆厂而不是闲置它们的公司来说，这几乎是不可想象的情况。

Loss of key customers and orders.

关键客户和订单的流失。

A foundry's success lives and dies by the big customers it can win and keep.

代工厂的成败取决于它能否赢得并留住大客户。

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Unfortunately for Samsung, most of the marquee names have stayed loyal to TSMC, or only given Samsung's small bits of business.

不幸的是，对三星来说，大多数知名公司仍然忠于台积电，或者只给三星少量业务。

Apple, the crown jewel client for any foundry, hasn't used Samsung for main chip production since around 2015.

苹果是任何代工厂的皇冠客户，但自2015年左右以来，它就没有使用三星生产主要芯片。

After the iPhone 6 S generation, Apple shifted entirely to TSMC for its A-series chips and never looked back, thanks in part to those yield power advantages TSMC demonstrated.

在iPhone 6S世代之后，苹果完全转向台积电生产A系列芯片，并且再也没有回头，部分原因是台积电展示的良率和性能优势。

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In video, after dabbling with Samsung 8 nanometers for one GPU generation, returned to TSMC for its next-gen GPUs, where performance and efficiency were critical, especially for AI accelerators.

在视频中，某公司在尝试三星8纳米工艺一代GPU后，回归台积电生产下一代GPU，因为性能和效率对AI加速器至关重要。

AMD has been all-in with TSMC for CPUs and GPUs.

AMD的CPU和GPU完全依赖台积电。

Qualcomm has noted, moved its flagship SOCs back to TSMC.

高通已将其旗舰SOC转回台积电生产。

Essentially, Samsung failed to retain these key external customers for the most advanced nodes.

本质上，三星未能留住这些关键外部客户在最先进制程上的合作。

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The result is a double whammy.

结果雪上加霜。

TSMC's revenue soars with high volume orders, giving its scale and learning benefits, while Samsung's fabs run below capacity or on older, less profitable nodes, or mainly produce Samsung's own chips, one telling statistic.

台积电因大量订单收入飙升，获得规模和学习优势，而三星的晶圆厂产能不足或依赖老旧低利润制程，主要生产自家芯片，这一数据很能说明问题。

In the first quarter of 2025, Samsung foundry's sales fell to $2.893 billion, an 11% drop quarter on quarter.

2025年第一季度，三星代工销售额降至28.93亿美元，环比下降11%。

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Its global share slid under 8%.

其全球份额跌破8%。

Meanwhile, smaller competitor SMIC slightly grew its sales and share.

与此同时，较小的竞争对手中芯国际的销售额和份额略有增长。

The global foundry market itself was a bit soft.

全球代工市场本身略显疲软。

It shrank 5% that quarter.

该季度市场萎缩5%。

But Samsung underperformed even that, indicating lost business.

但三星表现更差，表明业务流失。

Industry analysts pinned some blame on decline in orders from key customers, which widened the gap between TSMC and Samsung in late 2024.

行业分析师将部分责任归咎于关键客户订单减少，这扩大了2024年底台积电与三星的差距。

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Basically, some orders that might have gone to Samsung didn't, whether due to technical qualms or geopolitical concerns, or the aforementioned trust issues.

基本上，一些本可能交给三星的订单未能达成，无论是由于技术疑虑、地缘政治担忧，还是前述的信任问题。

Customer trust and conflict of interest.

客户信任和利益冲突。

Perhaps the most intriguing reason Samsung struggles to attract certain clients is a structural one.

三星难以吸引某些客户的最有趣原因或许是结构性问题。

Samsung isn't a pure play foundry.

三星并非纯粹的代工厂。

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It's a vertically integrated giant that also designs its own chips.

它是垂直整合的巨头，还设计自家芯片。

The system LSI division makes X-NOS processors, image sensors, et cetera, and sells finished products like smartphones and TVs.

其系统LSI部门生产X-NOS处理器、图像传感器等，并销售智能手机和电视等成品。

For some big tech companies, this is a red flag.

对一些大型科技公司而言，这是危险信号。

If your Apple or Qualcomm, you might worry that handing your latest chip designs to Samsung foundry could risk intellectual property leaks to Samsung's chip design teams, or give Samsung's end products a competitive insight.

如果你是苹果或高通，可能会担心将最新芯片设计交给三星代工可能导致知识产权泄露给三星芯片设计团队，或让三星的终端产品获得竞争洞察。

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This concern isn't just hypothetical.

这种担忧并非假设。

It's been voiced enough that insiders say it's a major factor behind Samsung's foundry, order drought.

业内人士多次表示，这是三星代工订单荒的主因。

Industry experts note that clients fear their design know-how might leak under Samsung's one-roof design and production setup.

行业专家指出，客户担心设计技术可能在三星设计与生产一体化的架构下泄露。

TSMC, by contrast, only does manufacturing and has built a reputation for strict secrecy and not competing with its customers.

相比之下，台积电仅从事制造，并以严格保密和不与客户竞争著称。

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Samsung has tried to assure clients of firewalls between business units, but the perception remains.

三星试图向客户保证业务部门间的防火墙，但疑虑仍在。

In May 2025, talk of spinning off Samsung's foundry business resurfaced as a possible solution to this conflict of interest.

2025年5月，分拆三星代工业务的讨论再次出现，作为解决利益冲突的可能方案。

Some analysts argue that separating the foundry unit and even potentially listing it as an independent company in the US could both remove the conflict concern and allow that unit to raise capital more freely.

一些分析师认为，分拆代工部门甚至可能将其作为独立公司在美国上市，既能消除冲突担忧，又能让该部门更自由融资。

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So far, it's just speculation, but it shows how serious the trust gap is viewed in the industry.

目前仅是猜测，但表明行业如何看待信任缺口的严重性。

Competitor moves.

竞争对手的动向。

Samsung's woes are exacerbated because its rivals have executed so well.

三星的困境因竞争对手表现出色而加剧。

TSMC's whole business is built around servicing others, and it has relentlessly kept on schedule or only slightly behind in each new generation.

台积电整个业务围绕服务他人构建，且每一代新技术都能严格按时或仅略微延迟交付。

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TSMC's ability to invest huge sums over $30 billion per year in capital expenditure recently into new capacity and R&D with guaranteed demand from Apple and others makes it hard to catch.

台积电每年投入超300亿美元资本支出用于新产能和研发，并有苹果等公司的需求保障，令对手难以追赶。

Even Intel, another giant trying to do foundry, has stumbled technologically in recent years, and Samsung had an opening to perhaps snatch Intel's foundry customers.

就连另一家尝试代工的巨头英特尔近年技术受挫，三星本有机会抢夺其客户。

But Intel has embarked on its own turnaround plan, even it separated its foundry arm internally.

但英特尔已启动转型计划，甚至内部拆分代工部门。

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Meanwhile, as mentioned, SMIC in China has been rising on a different front, focusing on mature, cheaper process nodes and domestic Chinese clients.

同时如前所述，中芯国际在另一领域崛起，专注于成熟廉价制程和中国国内客户。

SMIC is backed by Chinese government subsidies and willing to offer low prices, which helped it boost its market share from 5.5% to 6% in a single quarter, even amid global downturn.

中芯国际获中国政府补贴并愿提供低价，助其在一个季度内将份额从5.5%提升至6%，即便全球市场低迷。

Samsung competes with SMIC more on older node contracts for things like display driver chips, microcontrollers, et cetera, and losing share there hurts volume.

三星与中芯国际主要在显示驱动芯片、微控制器等老旧制程合同上竞争，份额流失影响产量。

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SMIC even managed to produce chips at an equivalent of seven nanometers, despite US sanctions, and is reportedly developing five nanometer-class tech, which shows their inching into advanced territory, too.

中芯国际甚至在美国制裁下实现等效7纳米芯片生产，据称正开发5纳米级技术，显示其也在进军先进领域。

Samsung is thus squeezed, TSMC taking the high-end SMIC and others like global foundries, UMC, undercutting on the low-end, and even Intel aiming to poach the remaining big clients in the middle.

三星因此被挤压——台积电占据高端，中芯国际、格芯和联电等低价竞争低端，英特尔则瞄准剩余中端大客户。

These factors together paint a picture of Samsung foundries worst-case scenario starting to unfold.

这些因素共同描绘出三星代工最坏情况开始显现的画面。

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In fact, an article from mid-2025 used exactly that phrasing, noting Samsung's steady decline and SMIC's ascent.

事实上，2025年中期一篇文章用此措辞，指出三星持续衰落而中芯国际崛起。

One professor was quoted saying, Samsung's foundry is in a tight spot.

一位教授称："三星代工处境艰难。"

It needs to step up its design and tech game and focus more on what customers want.

需提升设计和技术实力，更关注客户需求。

This encapsulates it.

这概括了问题。

Samsung must rebuild trust and technological parity to stop the bleeding.

三星必须重建信任和技术对等以止血。

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The company is aware of the issue.

公司意识到问题。

Reports say they've even reassigned some foundry engineers to the memory division and shook up leadership, reflecting internal recognition that things aren't working.

据报道，其甚至将部分代工工程师调至存储部门并改组领导层，反映内部承认现状不佳。

Before we move on, it's worth noting Samsung isn't giving up, far from it.

在继续前，值得注意三星并未放弃，远非如此。

They continue to invest heavily in new fabs, a huge new fab in Taylor, Texas is under construction, aiming at two nanometer production in a couple years.

其继续重资新建晶圆厂，德州泰勒一座大型新厂正在建设，目标几年内实现2纳米生产。

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And speaking of two nanometers, Samsung hopes that node will be a comeback moment.

谈到2纳米，三星希望该制程成为翻盘时刻。

Early rumors from late 2024 suggested Samsung's two nanometer development, called SF2, was going better than expected, with trial production yields around 20% to 30%.

2024年底早期传闻称，三星2纳米制程SF2开发超预期，试产良率达20%-30%。

If they can truly start two nanometer mass production by late 2025 or 2026 with competitive quality, Samsung might woo back some customers, especially those looking to diversify beyond TSMC, given geopolitical worries over Taiwan.

若能在2025年底或2026年以有竞争力的质量实现2纳米量产，三星或能赢回部分客户，尤其是因台湾地缘政治担忧寻求台积电替代的客户。

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So the foundry race isn't over yet, but right now, TSMC is far ahead and widening the gap, and Samsung is in scramble mode just to hold on to second place.

因此代工竞赛尚未结束，但目前台积电遥遥领先且扩大优势，三星正竭力保住第二。

Now let's turn to Samsung's other pillar, the memory business.

现在转向三星另一支柱——存储业务。

If losing ground in foundry is surprising, what happened in memory is almost shocking.

若在代工领域失利令人惊讶，存储业务的情况几乎令人震惊。

For over 30 years, Samsung led the world in memory chip sales.

30多年来，三星一直引领全球存储芯片销售。

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Samsung's prowess in DRAM, the main memory in computers, phones, servers, and NAND flash storage memory, was unmatched.

三星在DRAM（计算机、手机、服务器主存）和NAND闪存上的实力无可匹敌。

It often held around 40% to 50% market share in DRAM and 30% plus in NAND, reaping big profits during demand upswings.

其DRAM份额常达40%-50%，NAND超30%，需求高涨时获利丰厚。

Competitors like SK Heinex, also from South Korea, and Micron, USA, were always number two and number three respectively, formidable, but usually a step behind Samsung in size and technology.

来自韩国的SK海力士和美国美光等竞争对手分列二三位，实力强劲但规模和技术通常稍逊三星。

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That's why it made headlines in 2025 when SK Heinex overtook Samsung to become the world's largest DRAM vendor by revenue.

因此2025年SK海力士营收超越三星成为全球最大DRAM供应商时登上头条。

This was the first time in history, Samsung had lost the number one spot in memory since it became dominant decades ago.

这是三星数十年来主导存储市场后首次失去第一。

It's a seismic shift in the memory chip landscape.

这是存储芯片格局的地震级变化。

Let's unpack the data.

分析数据。

In the first quarter of 2025, SK Heinex captured 36% of global DRAM revenue, edging out Samsung's 34% share.

2025年第一季度，SK海力士占据全球DRAM营收的36%，略超三星的34%。

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US-based Micron was third with about 25%.

美国美光以约25%位列第三。

Just a quarter or two prior, Samsung was still number one, so this change happened fast.

仅一两季度前三星仍是第一，因此变化迅速。

What caused SK Heinex to surge past Samsung in memory?

什么导致SK海力士在存储领域超越三星?

In one word, AI.

一言以蔽之：AI。

The year 2024 saw an explosive growth in artificial intelligence workloads, especially with large language models and generative AI requiring powerful hardware.

2024年人工智能工作负载爆发式增长，尤其是大语言模型和生成式AI需要强大硬件。

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This translated to a spike in demand for a special type of memory called HBM, high bandwidth memory.

这转化为对高带宽内存HBM的特殊需求激增。

HBM is a kind of DRAM that is super fast and built in a 3D stacked way.

HBM是一种超快且以3D堆叠方式构建的DRAM。

Multiple layers of memory stacked vertically, connected by tiny wires through the silicon.

多层内存垂直堆叠，通过硅中微小导线连接。

It sits right next to processors on advanced packages, allowing massive data throughput, exactly what AI chips like GPUs and accelerators need.

其紧邻处理器置于先进封装上，实现海量数据吞吐，正是GPU和加速器等AI芯片所需。

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And here's the kicker, SK Heinex dominated the HBM market.

关键是SK海力士主导了HBM市场。

By 2025, SK Heinex held about 70% share in HBM sales, supplying memory for marquee AI products like Nvidia's A-100 and H-100 data center GPUs.

至2025年，SK海力士占据HBM销售约70%份额，为英伟达A100和H100数据中心GPU等明星AI产品供应内存。

In fact, SK Heinex was the first to develop HBM technology back in 2013 and persisted with it even when demand was low.

事实上，SK海力士早在2013年就率先开发HBM技术，并在需求低迷时坚持投入。

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Now that demand has skyrocketed, Heinex was ready, producing 5th generation HBM 3E chips and reportedly working on the next gen HBM 4.

如今需求飙升，海力士已做好准备，生产第五代HBM3E芯片并据称研发下一代HBM4。

This foresight paid off tremendously in 2018, 2024 to 2025, as every AI server wanted more HBM.

这一远见在2018、2024至2025年获得巨大回报，因每台AI服务器都需要更多HBM。

Nvidia, which practically cornered the AI accelerator market, was buying tons of HBM from Heinex and to a lesser extent micron.

几乎垄断AI加速器市场的英伟达从海力士大量采购HBM，少量来自美光。

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Samsung, on the other hand, fell behind in this niche but critical segment.

而三星在这一细分但关键的领域落后。

Samsung did produce HBM 2.

三星确实生产过HBM2。

They had earlier HBM 2, HBM 2E products.

其早期有HBM2、HBM2E产品。

But when it came to the latest HBM 3-slash 3E generation, Samsung lagged in time to market and volume.

但在最新HBM3/3E世代，三星上市时间和产量均落后。

In fact, Samsung publicly acknowledged delays in ramping up its newest HBM chips.

事实上，三星公开承认最新HBM芯片量产延迟。

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Their HBM 3E was late to reach customers, which meant Samsung missed out on much of the 2023 to 2024 AI boom.

其HBM3E交付客户较晚，意味着三星错失2023-2024年大部分AI热潮。

By the time Samsung was readying its high-end memory for AI, SK Heinex had already locked in supply deals and was shipping in volume.

当三星准备高端AI内存时，SK海力士已锁定供应协议并大量出货。

A Bloomberg piece starkly noted that Samsung's hope of supplying Nvidia with HBM was snuffed out by its delays, writers SK Heinex began volume production.

彭博社文章尖锐指出，三星因延迟供货英伟达HBM的希望破灭，而SK海力士已开始量产。

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Micron also started shipping HBM products, meaning Samsung suddenly found itself third in a race it didn't expect to lose.

美光也开始出货HBM产品，意味着三星突然在这场本以为不会输的竞赛中跌至第三。

The result, SK Heinex's revenues and profits surged.

结果SK海力士收入和利润激增。

Heinex actually reported a record operating profit of 7 trillion one in Q3 2024 on the back of AI memory demand.

海力士2024年第三季度因AI内存需求创下7.1万亿韩元运营利润纪录。

A dramatic rebound from what had been a brutal memory downturn earlier that year.

较当年早些时候存储业严峻低迷大幅反弹。

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Samsung, meanwhile, saw far more modest results, even swinging to losses in its memory division during the worst of the slump.

而三星业绩平淡得多，甚至在最严重低迷期存储部门出现亏损。

Now, it's important to note that memory chips are a famously cyclical business.

需注意存储芯片是著名的周期性行业。

The industry goes through boom and bust cycles.

该行业经历繁荣与萧条周期。

Shortages and high prices lead manufacturers to expand supply.

短缺和高价促使制造商扩大供应。

Which then causes over supply and price crashes and the cycle repeats.

这进而导致供应过剩和价格暴跌，循环往复。

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Samsung has navigated these cycles for decades.

三星数十年来一直应对这些周期。

Sometimes by not cutting production in downturns to squeeze out weaker rivals.

有时通过在低迷期不减产来挤压较弱对手。

Using its deep pockets to weather losses.

利用雄厚资金承受亏损。

In 2022 to 2023, a severe chip glut hit has demand for PCs and phones fell post-pandemic.

2022至2023年，后疫情时代PC和手机需求下降导致芯片严重过剩。

Samsung initially kept output high.

三星最初保持高产量。

Causing it to post a historic loss in chips, its worst since 2009.

导致其芯片业务出现2009年以来最严重亏损。

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And inventory to balloon.

库存激增。

It finally agreed to cut memory production in 2023 as losses mounted.

随着亏损加剧，三星最终在2023年同意削减内存产量。

SK Hynix and Micron had preemptively cut theirs to stabilize the market.

SK海力士和美光已提前减产以稳定市场。

So one could argue Samsung's strategy misstep started there.

因此可以说三星的战略失误始于此处。

They slept on how quickly AI would become the only growth driver.

他们忽视了AI迅速成为唯一增长动力的趋势。

And how the old playbook of flooding the market was backfiring.

以及过去用产能淹没市场的策略正在适得其反。

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By late 2023, however, all eyes were on AI-driven memory demand.

但到2023年底，焦点已转向AI驱动的内存需求。

Which pulled the memory market out of the gutter faster than expected.

这比预期更快地将内存市场拉出低谷。

Analogy.

打个比方。

It's as if Samsung prepared for a long winter.

就像三星为漫长寒冬做了准备。

But spring, AI demand, arrived early.

但春天（AI需求）提前到来。

And Samsung was caught without the seeds, the latest HBM fully ready to plant.

三星却未备好种子（最新HBM技术）播种。

SK Hynix was already harvesting.

SK海力士已在收割。

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This translated into market share flip in Dram.

这导致DRAM市场份额翻转。

Because HBM, though a small portion of total bits, is very high value, expensive per chip.

因为HBM虽只占存储总量一小部分，但单价极高。

So revenue share shifted strongly in Hynix's favor due to product mix.

因此产品组合使收入份额大幅倾向海力士。

It's not that Samsung's memory technology became bad overall.

并非三星的内存技术整体变差。

In commodity D-RAM and NAND, Samsung still has very advanced processes.

在传统DRAM和NAND领域，三星仍拥有先进工艺。

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They continued to lead in NAND flash layering.

他们持续引领NAND闪存堆叠技术。

Like being first to mass produce 128 layer and 200 layer 3D NAND, etc.

例如率先量产128层和200层3D NAND。

In standard D-RAM for PCs and phones, DDR4, DDR5, LPDDR5, etc. Samsung is competitive.

在PC和手机用的标准DRAM（如DDR4、DDR5、LPDDR5）上，三星仍有竞争力。

The issue was more missing the boat on the hottest growth segment, HBM.

问题更多是错过了最热增长领域HBM。

And possibly being too slow to pivot.

且转型可能过慢。

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Interestingly, SK Hynix's bold bet on HBM highlights a cultural difference.

有趣的是，SK海力士对HBM的大胆押注凸显文化差异。

A smaller player taking a risk on a niche tech.

小公司敢于押注小众技术。

Versus the giant Samsung focusing on volume products.

而巨头三星专注于量产产品。

When that niche became the next big thing.

当小众技术变成新风口时。

The smaller player reaped outsized rewards.

小公司获得了超额回报。

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